

RESEARCH PAPER

**PRODUCT LINE PROFITABILITY & MARGIN
PERFORMANCE ANALYSIS
NASSAU CANDY DISTRIBUTOR**

An Internship Project Report

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September 2026

1. Introduction

Profitability analysis is important for businesses to understand which products and divisions contribute most to revenue and profit. This project focuses on analyzing the product line profitability and margin performance of Nassau Candy Distributor.

The analysis examines sales, cost, gross profit, gross margin, units sold and profit contribution at the product and division levels. It also identifies products with high profitability, products with margin risks and areas where costs may be affecting profitability.

The project uses data analysis and visualization techniques to generate meaningful business insights and support data-driven decision-making.

2. Problem Statement

Nassau Candy Distributor needs to understand which products and divisions are driving profitability and which products have weak margins or high costs. High sales do not always indicate high profitability, making it important to evaluate sales, cost, gross profit and margin performance together.

This project addresses the need for a detailed profitability analysis to identify high-performing products, margin-risk products, profit concentration, division-level performance and cost-related issues. The analysis aims to provide meaningful insights that can support better pricing, cost management and product-level business decisions.

3. Objectives

- To analyze product-level sales, cost and gross profit.
- To calculate **Gross Margin %** and **Profit per Unit** for each product.
- To identify high-performing products and products with weak profit margins.
- To compare sales and profitability across different divisions.
- To analyze revenue and profit concentration using Pareto analysis.
- To identify products with high costs relative to their sales.
- To highlight products that require margin-risk attention.
- To develop an interactive Streamlit dashboard for profitability analysis and business insights.

4. Dataset Description

The dataset used in this project contains **10,194 records and 22 columns** related to orders, products, customers, sales, costs, shipping, divisions and regions.

The dataset contains information such as Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit and Cost.

For the profitability analysis, the main variables used were **Order Date, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit and Cost**. These variables were used to calculate profitability metrics and evaluate product and division performance.

5. Tools and Technologies

The project was carried out using data analysis, visualization and business intelligence tools.

The following technologies were used:

- **Python** – Used for data cleaning, transformation, calculations, and analysis.
- **Pandas** – Used for data manipulation, aggregation.
- **Jupyter Notebook** – Used for performing and documenting the data analysis.
- **Streamlit** – Used to develop the interactive profitability dashboard.

6. Methodology

The project followed a systematic data analysis approach to evaluate product profitability and margin performance. The methodology involved data cleaning, financial validation, metric calculation, product-level analysis, division-level analysis, Pareto analysis, cost and margin diagnostics and dashboard development.

Step 1: Data Collection and Understanding

The dataset was examined to understand its structure, variables, data types and business context.

Step 2: Data Cleaning and Validation

The data was checked for missing values, duplicate records, incorrect data types and inconsistencies. Financial values were also validated by comparing Gross Profit with Sales minus Cost.

Step 3: Profitability Metrics

Key metrics such as Gross Margin %, Profit per Unit, Revenue Contribution % and Profit Contribution % were calculated to evaluate product performance.

Step 4: Product and Division Analysis

Products were aggregated and ranked based on sales, gross profit and margin. Division-level performance was analyzed to identify major contributors to revenue and profitability.

Step 5: Pareto Analysis

Pareto analysis was performed to determine the concentration of revenue and gross profit among the products.

Step 6: Cost and Margin Diagnostics

Cost as a percentage of sales was analyzed to identify products with high cost levels and weak margins. Margin risk flags were also used to highlight products requiring further attention.

Step 7: Dashboard Development

An interactive Streamlit dashboard was developed with filters for order date, division, minimum gross margin and product search. The dashboard presents product profitability, division performance, cost and margin diagnostics and profit concentration.

7. Data Cleaning & Validation

Before performing the profitability analysis, the dataset was examined and cleaned to ensure data quality and consistency. The dataset contained **10,194 records and 22 columns**.

The following validation checks were performed:

1. **Missing Values:** The dataset was checked for missing values across all columns. No missing values were found.
2. **Duplicate Records:** The dataset was checked for duplicate records. No duplicate records were identified.
3. **Data Types:** Date and numerical columns were reviewed and converted into appropriate data types where required.
4. **Financial Validation:** Gross Profit was validated against the relationship:

Gross Profit = Sales – Cost

The calculated values matched the existing Gross Profit values, confirming the consistency of the financial data.

These checks ensured that the dataset was suitable for further profitability analysis and visualization.

8. Profitability Analysis

Profitability analysis was performed at the product level to understand the relationship between sales, cost, gross profit, and profit margins. The analysis also considered profit generated per unit and the contribution of individual products to total revenue and profit.

8.1 Key Profitability Metrics

The following metrics were calculated to evaluate product profitability:

Gross Margin % measures the percentage of sales retained as gross profit after accounting for product cost.

$$\text{Gross Margin \%} = (\text{Gross Profit} / \text{Sales}) \times 100$$

Profit per Unit measures the gross profit generated from each unit sold.

$$\text{Profit per Unit} = \text{Gross Profit} / \text{Units}$$

Revenue Contribution % measures the percentage of total revenue generated by each product.

Profit Contribution % measures the percentage of total gross profit contributed by each product.

8.2 Product-Level Findings

The product-level analysis showed that the Chocolate division was the major contributor to overall sales and gross profit. The five highest-sales products were all Wonka Bar products and also represented the five highest gross-profit products.

The top five products by sales were:

1. **Wonka Bar - Triple Dazzle Caramel** – Sales: **\$28,485.00**, Gross Profit: **\$18,610.20**, Gross Margin: **65.33%**

2. **Wonka Bar - Scrumdiddlyumptious** – Sales: **\$27,874.80**, Gross Profit: **\$19,357.50**, Gross Margin: **69.44%**
3. **Wonka Bar - Milk Chocolate** – Sales: **\$26,867.75**, Gross Profit: **\$17,443.37**, Gross Margin: **64.92%**
4. **Wonka Bar - Fudge Mallows** – Sales: **\$24,890.40**, Gross Profit: **\$16,593.60**, Gross Margin: **66.67%**
5. **Wonka Bar - Nutty Crunch Surprise** – Sales: **\$23,574.95**, Gross Profit: **\$16,819.95**, Gross Margin: **71.35%**

Among these products, **Wonka Bar - Nutty Crunch Surprise** had the highest gross margin at **71.35%**, while **Wonka Bar - Scrumdiddlyumptious** generated the highest gross profit of **\$19,357.50**.

The analysis indicates that these core Chocolate products are important contributors to the company's profitability and should receive continued attention in terms of inventory, pricing, and product availability.

8.3 Product Profitability Classification

To further evaluate product performance, products were classified using their **Sales** and **Gross Margin %** relative to the median values. The median sales value was **597.50**, while the median gross margin was **62.31%**.

Based on these measures, products were divided into four categories:

Category	Number of Products	Interpretation
Star Product	5	High sales and high margin
Growth Opportunity	3	Lower sales but high margin
Margin Risk	3	High sales but lower margin
Low Priority	4	Lower sales and lower margin

This classification helps distinguish products that are strong contributors from products that may require improvement or further evaluation.

8.4 Division Performance

The division-level analysis was conducted to compare sales, units sold, cost, gross profit, and gross margin across the three divisions: **Chocolate, Sugar, and Other**.

Division	Sales	Gross Profit	Gross Margin%	Revenue Contribution %	Profit Contribution %
Chocolate	\$131,692.90	\$88,824.62	67.45%	92.88%	95.06%
Other	\$9,663.25	\$4,333.45	44.84%	6.82%	4.64%
Sugar	\$427.48	\$284.73	66.61%	0.30%	0.30%

Key observations:

- The **Chocolate division** is the primary contributor, generating **92.88% of total revenue and 95.06% of total gross profit**.
- The **Other division** has a considerably lower gross margin of **44.84%**, and its profit contribution is lower than its revenue contribution.
- The **Sugar division** has a relatively strong gross margin of **66.61%**, but its overall sales volume and contribution are very small.

Overall, the results show that profitability is highly concentrated in the Chocolate division, while the Other division requires greater attention to cost and margin performance.

8.5 Cost vs. Margin Diagnostics

Cost-to-sales analysis was performed to identify products where a high proportion of sales was consumed by product cost. The **Cost % of Sales** metric was calculated as:

$$\text{Cost \% of Sales} = (\text{Cost} / \text{Sales}) \times 100$$

The analysis showed significant differences in cost efficiency across products. **Kazookles** had the highest cost burden, with cost representing **92.31% of sales**, resulting in a gross margin of only **7.69%**. This indicates a significant margin concern.

Lickable Wallpaper had a cost-to-sales ratio of **50.00%** and a gross margin of **50.00%**, while **Wonka Gum** had a cost-to-sales ratio of **48.00%** and a gross margin of **52.00%**.

These results indicate that products with relatively high costs compared with their sales can reduce overall profitability. Such products may require further review of pricing, sourcing costs, or product-level expenses.

Key Finding:

Kazookles represents the most significant cost-related margin concern because its cost accounts for more than 90% of sales. Lickable Wallpaper and Wonka Gum also warrant

attention because their margins are considerably lower than the core high-performing Chocolate products.

8.6 Margin Risk Analysis

Products were evaluated based on their sales level and gross margin in order to identify products that may require management attention. Products with gross margins below the median margin of **62.31%** were examined in greater detail.

Three products were identified as **High Priority** based on relatively high sales combined with below-median gross margins:

- **Kazookles** – Gross Margin: **7.69%**
- **Lickable Wallpaper** – Gross Margin: **50.00%**
- **Wonka Gum** – Gross Margin: **52.00%**

Four additional products were classified as **Monitor** because they had below-median margins but relatively lower sales:

Fun Dip, Nerds, SweeTARTS, and Fizzy Lifting Drinks.

The remaining products were classified as **Healthy** based on the margin and sales criteria used in the analysis.

This classification helps prioritize management attention. High-priority products may require further investigation into pricing, sourcing costs, or other factors affecting their margins.

8.7 Profit Concentration / Pareto Analysis

Pareto analysis was performed to understand how gross profit is distributed across the product portfolio. Products were ranked from highest to lowest based on gross profit, and their cumulative profit contribution was calculated.

The analysis showed a high concentration of profitability among a small number of products. **Only 5 out of 15 products, representing approximately 33.33% of the product portfolio, generated 95.06% of total gross profit.**

The same five products also accounted for **92.88% of total revenue**. These products were:

- Wonka Bar - Scrumdiddlyumptious
- Wonka Bar - Triple Dazzle Caramel
- Wonka Bar - Milk Chocolate

- Wonka Bar - Nutty Crunch Surprise
- Wonka Bar - Fudge Mallows

This finding indicates that the company's revenue and profitability are highly concentrated among a small group of core products. Maintaining the availability and profitability of these products is therefore important for overall business performance.

8.8 Regional Performance

Regional analysis was performed to compare sales, gross profit, and gross margin across the four regions: Atlantic, Gulf, Interior, and Pacific.

The **Pacific region** generated the highest sales of **\$46,301.53** and the highest gross profit of **\$30,485.94**. The **Interior region** recorded the highest gross margin at **66.43%**.

The gross margins across the four regions were relatively close, ranging from **65.47% to 66.43%**. This indicates that regional differences in margin performance were relatively small compared with the differences observed at the product level.

Therefore, the analysis suggests that **product-level profitability and cost structure are more important areas of investigation than regional margin differences** in this dataset.

8.9 Monthly Sales and Profit Trend

Monthly sales and gross profit were analyzed to identify changes in business performance over time. The analysis covered the period from **January 2024 to December 2025**.

The results showed an overall improvement in sales and gross profit during **2025 compared with 2024**. Performance was particularly strong during the final quarter of both years, with sales increasing significantly in September, November, and December.

December 2025 recorded the highest monthly sales of \$12,338.27 and the highest gross profit of \$8,115.95 during the analyzed period. Gross profit generally increased along with sales, without a clear indication of a major profitability decline during the highest-sales months.

The monthly trend therefore indicates stronger business performance in 2025, particularly toward the end of the year.

9. Key Findings

The analysis of the Nassau Candy Distributor dataset produced the following key findings:

1. **Strong Chocolate Division Contribution:** The Chocolate division generated **92.88% of total revenue and 95.06% of total gross profit**, making it the major contributor to overall profitability.
2. **High Profit Concentration:** Only **5 out of 15 products (33.33%)** contributed approximately **95.06% of total gross profit and 92.88% of total revenue**.
3. **Top Product Performance:** Wonka Bar - Scrumdiddlyumptious generated the highest gross profit at **\$19,357.50**, while Wonka Bar - Nutty Crunch Surprise recorded the highest gross margin among the top five products at **71.35%**.
4. **Margin Risk Products:** Kazookles had a particularly low gross margin of **7.69%**, with cost representing **92.31% of sales**. Lickable Wallpaper and Wonka Gum also showed comparatively lower margins.
5. **Regional Performance:** Regional gross margins were relatively consistent, ranging from **65.47% to 66.43%**, suggesting that product-level factors had a greater impact on profitability than regional differences.
6. **Improved 2025 Performance:** Monthly analysis showed stronger sales and gross profit in **2025 compared with 2024**, with **December 2025** recording the highest monthly sales and gross profit.

10. Business Recommendations

Based on the profitability, cost, margin, division, and concentration analysis, the following recommendations can be considered:

1. **Focus on Core Profitable Products:** The five major Wonka Bar products contribute a large share of total revenue and gross profit. Their availability, pricing and inventory levels should be monitored carefully.
2. **Review High-Risk Products:** Products such as **Kazookles, Lickable Wallpaper and Wonka Gum** should be reviewed because of their comparatively lower gross margins. Pricing, sourcing costs and other product-related costs can be examined to identify opportunities for improvement.
3. **Improve Cost Efficiency:** Products with a high Cost % of Sales should be investigated to understand whether procurement costs, pricing, or other factors are affecting their margins.
4. **Monitor the Other Division:** The Other division has a lower gross margin of **44.84%** compared with the Chocolate division's **67.45%**. Its product-level performance should therefore be monitored to identify opportunities for margin improvement.
5. **Maintain a Balanced Product Portfolio:** Since profitability is highly concentrated among a small number of products, management can continue supporting the core products while evaluating lower-performing products for improvement, repositioning or rationalization.
6. **Use the Dashboard for Ongoing Monitoring:** The Streamlit dashboard can be used to interactively monitor product profitability, division performance, cost levels and profit concentration using different filters.

11. Streamlit Dashboard

An interactive Streamlit dashboard was developed to present the profitability analysis in an easy-to-use format. The dashboard allows users to explore product and division performance and examine cost and margin-related indicators.

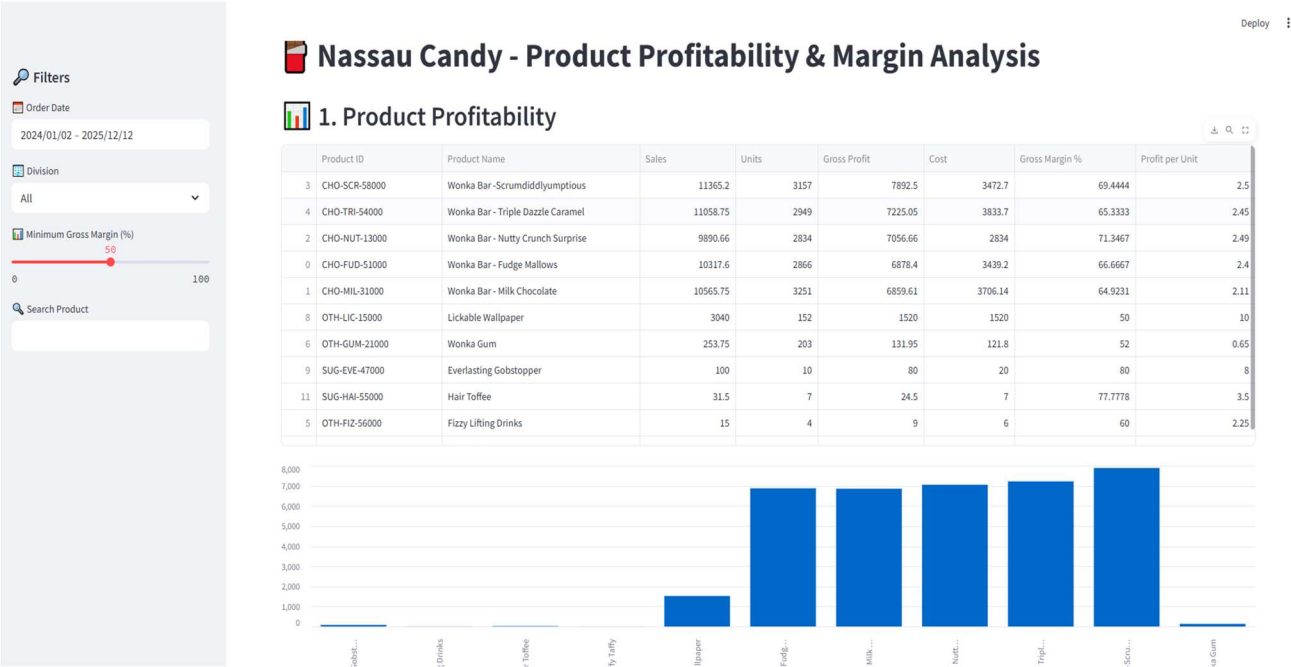
The dashboard includes the following features:

- **Date Range Filter:** Allows users to analyze performance for a selected period.
- **Division Filter:** Allows users to focus on a specific business division.
- **Minimum Gross Margin Filter:** Allows users to display products above a selected margin threshold.
- **Product Search:** Allows users to search for specific products.

The dashboard contains four major analytical modules:

1. **Product Profitability** – Displays product-level sales, units, cost, gross profit, gross margin, and profit per unit.
2. **Division Performance** – Compares sales, gross profit, and margin performance across divisions.
3. **Cost vs. Margin Diagnostics** – Highlights the relationship between product costs and gross margins.
4. **Profit Concentration / Pareto Analysis** – Shows cumulative profit contribution across products.

The dashboard provides an interactive way to examine the results of the analysis and supports more efficient exploration of product profitability and margin performance.



12. Conclusion

This project analyzed the product line profitability and margin performance of Nassau Candy Distributor using sales, cost, gross profit, units and margin-related metrics.

The analysis identified the Chocolate division as the major contributor to overall revenue and gross profit, while a small number of core products accounted for a large proportion of total profitability. The analysis also identified products with comparatively weak margins and high cost-to-sales ratios that may require further review.

Regional analysis showed relatively consistent margins across regions, while the monthly analysis indicated stronger sales and gross profit performance during 2025, particularly toward the end of the year.

An interactive Streamlit dashboard was developed to make the analysis accessible through filters and visualizations. Overall, the project demonstrates how data analysis can be used to evaluate product profitability, identify margin risks, understand profit concentration and support data-driven business decisions.